Digestive System: the GI map

Build this as we go. Every stop on the tube in order, and every organ that pours into it. Competencies 109 to 125. Name _____

THE TUBE: FOLLOW THE BITE

1. Mouth (oral cavity)
Epithelium: _____
Happens here: _____
Boundaries: _____

↓

2. Pharynx
Food uses: _____
Sealed off: _____
Gate out: _____

↓

3. Esophagus
Epithelium: _____
Outer layer: _____
Through the diaphragm at: _____
Gate out: _____

↓

4. Stomach
Regions in order: _____
Epithelium: _____
Extra muscle layer: _____
Folds: _____
Gate out: _____

↓

5. Small intestine
Duodenum, 25 cm: _____
Jejunum, 1 m: _____
Ileum, 2 m: _____
Surface: _____
Gate out: _____

↓

6. Large intestine
In order: _____
Only here: _____
Epithelium: _____
Job: _____

↓

7. Rectum and anal canal
Rectum: _____
Anal canal lining near the anus: _____
Two gates: _____

ACCESSORY ORGANS: WHAT POURS INTO THE TUBE, AND WHERE

Salivary glands, into the mouth
Parotid duct opens: _____
Submandibular duct opens: _____
Sublingual ducts open: _____

Teeth and tongue
Tongue muscles: _____
Papillae without taste buds: _____
Adult teeth: _____

Liver, into the duodenum by bile
Lobes: _____
Lobule: _____
Bile path: _____

Gallbladder
Regions: _____
Its duct: _____
Job: _____

Pancreas, into the duodenum
Regions: _____
Main duct joins the _____
Accessory duct opens at the _____
Acini (exocrine) and _____

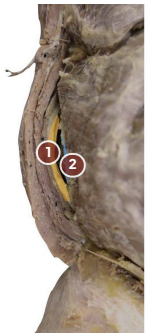
DRAW: THE WHOLE CANAL IN THE TORSO, DIAPHRAGM AT THE TOP, ACCESSORY ORGANS BESIDE IT, A RING AT EVERY SPHINCTER

Digestive System, Part I

Competencies 109 to 114, 120, 121: peritoneum and folds, retroperitoneal organs, the gut wall and its sphincters, mouth, salivary glands, tongue, esophagus.

Name _____

A. Peritoneum and the omenta (cadaver) COMPETENCY 111



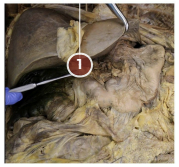
Cadaver, peritoneum: parietal and visceral layers

1 _____
2 _____



Cadaver, greater omentum

1 _____



Cadaver, lesser omentum

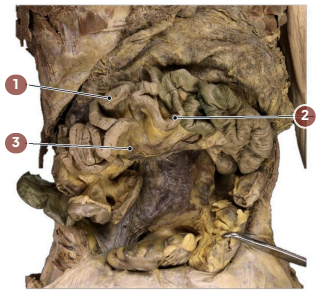
1 _____

FIVE FOLDS: WHAT EACH ONE CONNECTS

Greater omentum: _____
Lesser omentum: _____
Falciform ligament: _____
Mesentery: _____
Mesocolon: _____

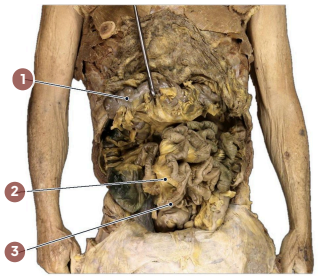
Extra notes and drawings

B. Mesentery, and what sits behind the peritoneum (cadaver) COMPETENCY 111, 112



Cadaver, mesentery of the small intestine

1 _____
2 _____
3 _____



Cadaver, transverse colon lifted

1 _____
2 _____
3 _____

FILL IN

Intraperitoneal: _____
Retroperitoneal: _____
Retroperitoneal organs: _____
On a mesocolon: _____

Extra notes and drawings

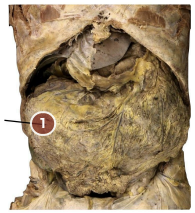
Digestive System, Part I STRUCTURE NOTES

A. Peritoneum and the omenta (cadaver) · Competency 111 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

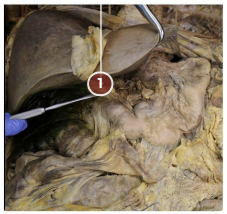
Name _____



Cadaver, peritoneum: parietal and visceral layers



Cadaver, greater omentum



Cadaver, lesser omentum

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

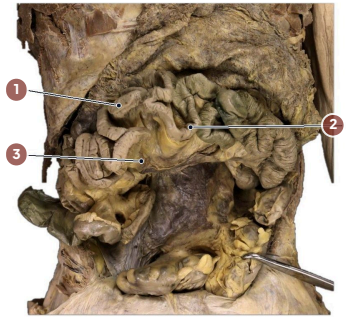
1 _____
Def _____
Function _____

1 _____
Def _____
Function _____

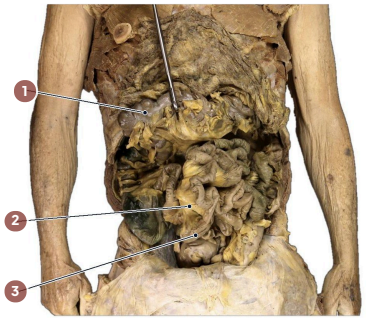
Digestive System, Part I STRUCTURE NOTES

B. Mesentery, and what sits behind the peritoneum (cadaver) · Competency 111, 112 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Cadaver, mesentery of the small intestine



Cadaver, transverse colon lifted

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

Digestive System, Part I

Competencies 109 to 114, 120, 121: peritoneum and folds, retroperitoneal organs, the gut wall and its sphincters, mouth, salivary glands, tongue, esophagus.

Name _____

C. The wall of the tube, and its gates (draw it) COMPETENCY 109, 110, 114

FOUR LAYERS, LUMEN OUTWARD

1. _____
2. _____
3. _____
4. _____

FILL IN

Submucosal plexus controls _____

Myenteric plexus controls _____

Stratified squamous lines the _____

Simple columnar lines the _____

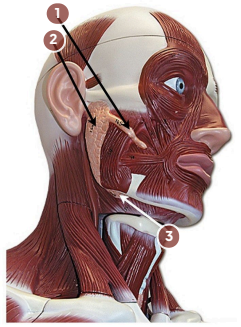
SIX SPHINCTERS, IN THE ORDER FOOD MEETS THEM

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

DRAW: A CROSS SECTION OF THE TUBE AS FOUR RINGS, WITH THE TWO PLEXUSES DOTTED IN

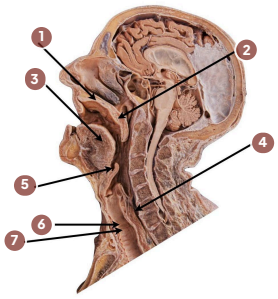
Extra notes and drawings

D. Mouth, salivary glands, tongue, esophagus COMPETENCY 113, 120, 121



Model, salivary glands

1. _____
2. _____
3. _____



Cadaver, midsagittal: palate, tongue, esophagus

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

FILL IN

Fauces: _____

Parotid duct: _____

Submandibular duct: _____

Sublingual ducts: _____

Lingual glands: _____

No taste buds: _____

Extrinsic muscles: _____

Intrinsic muscles: _____

Esophageal hiatus: _____

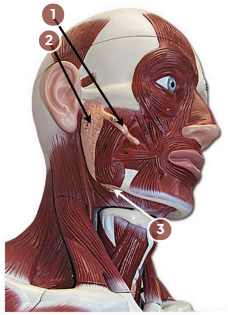
Extra notes and drawings

Dr. Sharilyn Rennie · BIO 004 Human Anatomy · Solano Community College · Photos: BIO 004 Digital Atlas, drsrennie-stack.github.io/new-digital-atlas

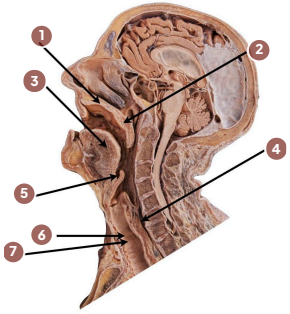
Digestive System, Part I STRUCTURE NOTES

D. Mouth, salivary glands, tongue, esophagus · Competency 113, 120, 121 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Model, salivary glands



Cadaver, midsagittal: palate, tongue, esophagus

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

1 _____
Def _____
Contributing structures _____
Borders _____
Blood supply (info only) .....
Function _____
Clinical _____

2 _____
Def _____
Made of _____
Borders _____
Action _____
Nerve _____
Clinical _____

3 _____
Def _____
Made of _____
Parts _____
Nerve _____
Blood supply (info only) .....
Clinical _____

4 _____
Def _____
Location _____
Lining _____
Nerve _____
Blood supply (info only) .....
Function _____
Clinical _____

5 _____
Def _____
Attaches _____
Action _____
Nerve _____
Clinical _____

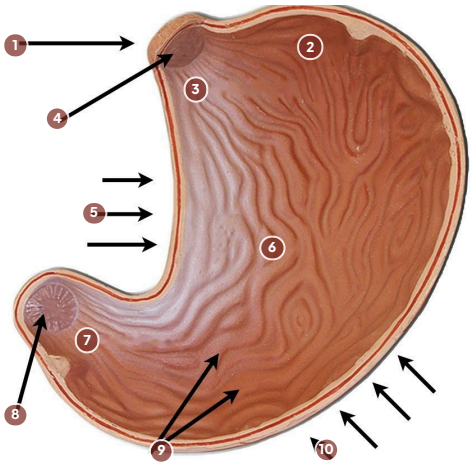
6 _____
Def _____
Function _____
Note _____

7 _____
Def _____
Contributing structures _____
Nerve _____
Function _____
Clinical _____

Digestive System, Part II

Competencies 115 and 122 to 125: stomach, liver, gallbladder and bile path, pancreas, and the accessory organ disorders. Name _____

A. Stomach regions and features (model) COMPETENCY 115



Model, stomach, interior

1 _____ 6 _____
2 _____ 7 _____
3 _____ 8 _____
4 _____ 9 _____
5 _____ 10 _____

FILL IN

Muscle layers, inside out: _____

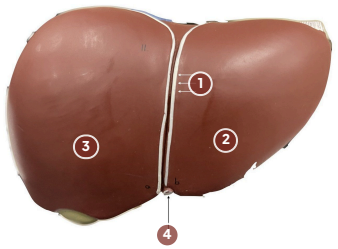
Extra layer: _____

Lesser curvature faces _____

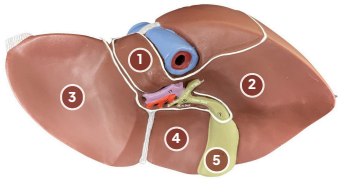
Greater curvature holds the _____

Rugae are folds of _____

B. Liver lobes and ligaments (model) COMPETENCY 122



Model, liver, anterior



Model, liver, inferior

FILL IN

From the front: _____

From below: _____

Falciform ligament reaches the _____

Ligamentum teres was the _____

Porta hepatis, in: _____

Porta hepatis, out: _____

1 _____
2 _____
3 _____
4 _____

1 _____
2 _____
3 _____
4 _____
5 _____

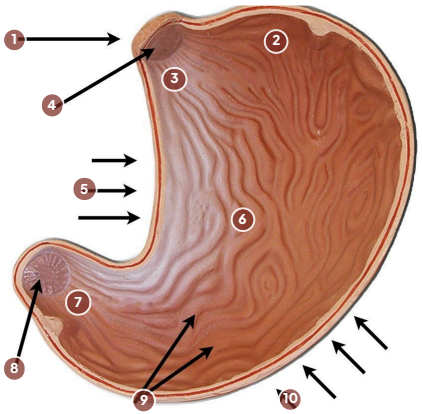
Extra notes and drawings

Extra notes and drawings

Digestive System, Part II STRUCTURE NOTES

A. Stomach regions and features (model) · Competency 115 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Model, stomach, interior

1 _____
Def _____
Location _____
Lining _____
Nerve _____
Blood supply (info only) .....
Function _____
Clinical _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

5 _____
Def _____
Function _____

6 _____
Def _____
Function _____

7 _____
Def _____
Function _____

8 _____
Def _____
Function _____

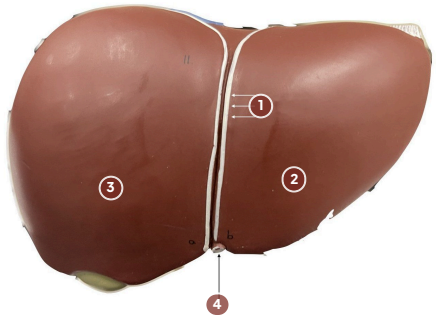
9 _____
Def _____
Function _____

10 _____
Def _____
Function _____

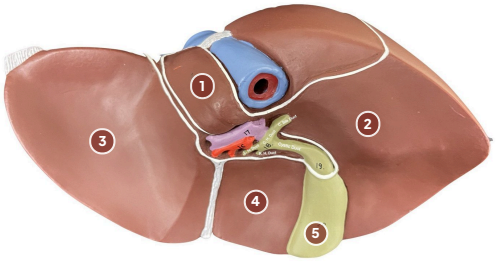
Digestive System, Part II STRUCTURE NOTES

B. Liver lobes and ligaments (model) · Competency 122 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Model, liver, anterior



Model, liver, inferior

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Location _____
Blood supply (info only) .....
Note _____

4 _____
Def _____
Function _____

1 _____
Def _____
Function _____

2 _____
Def _____
Location _____
Blood supply (info only) .....
Note _____

3 _____
Def _____
Function _____

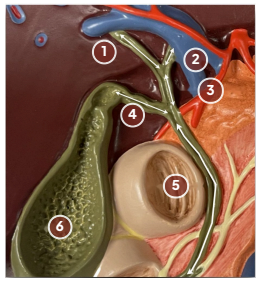
4 _____
Def _____
Function _____

5 _____
Def _____
Function _____

Digestive System, Part II

Competencies 115 and 122 to 125: stomach, liver, gallbladder and bile path, pancreas, and the accessory organ disorders. Name _____

C. Bile path and the pancreas (model and cadaver) COMPETENCY 123, 124



Model, gallbladder and the biliary tree

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

A DROP OF BILE, IN ORDER

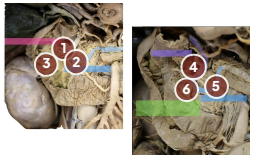
1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

FILL IN

Joins the common bile duct: _____

Exocrine: _____

Endocrine: _____

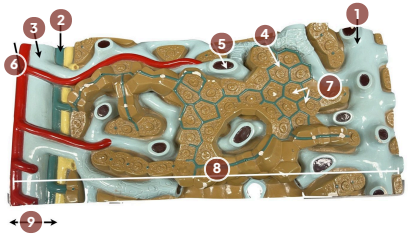


Cadaver, pancreatic and bile ducts entering the duodenum

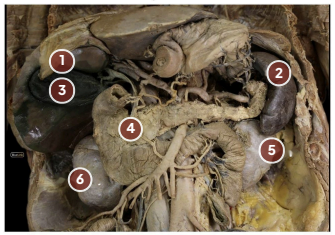
1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Extra notes and drawings

D. Liver lobule, and the organs in place (model and cadaver) COMPETENCY 122, 125



Model, liver lobule



Cadaver, major organs of the abdomen

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____

FILL IN

Blood flows toward the _____

Bile flows toward the _____

Portal triad: _____

DISORDERS: NAME THE ORGAN

Gallstones, cholecystitis: _____

Hepatitis, cirrhosis: _____

Pancreatitis: _____

Mumps: _____

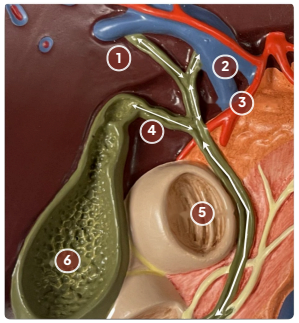
Stone at the ampulla blocks _____

Extra notes and drawings

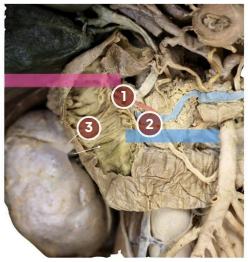
Digestive System, Part II STRUCTURE NOTES

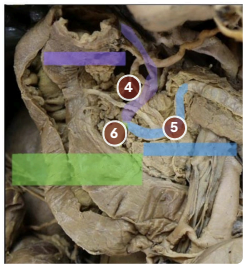
C. Bile path and the pancreas (model and cadaver) · Competency 123, 124 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Model, gallbladder and the biliary tree





Cadaver, pancreatic and bile ducts entering the duodenum

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

5 _____
Def _____
Function _____

6 _____
Def _____
Function _____

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

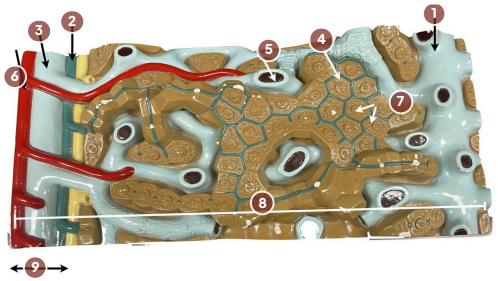
5 _____
Def _____
Function _____

6 _____
Def _____
Function _____

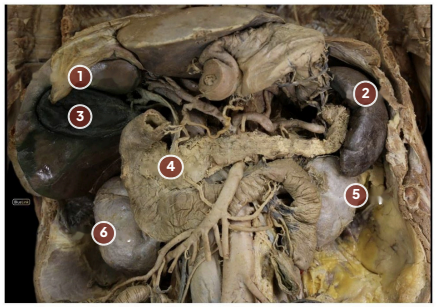
Digestive System, Part II STRUCTURE NOTES

D. Liver lobule, and the organs in place (model and cadaver) · Competency 122, 125 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Model, liver lobule



Cadaver, major organs of the abdomen

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

5 _____
Def _____
Function _____

6 _____
Def _____
Function _____

7 _____
Def _____
Function _____

8 _____
Def _____
Function _____

9 _____
Def _____
Function _____

1 _____
Def _____
Location _____
Blood supply (info only) .....
Note _____

2 _____
Def _____
Location _____
Borders _____
Parts _____
Blood supply (info only) .....
Function _____
Clinical _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

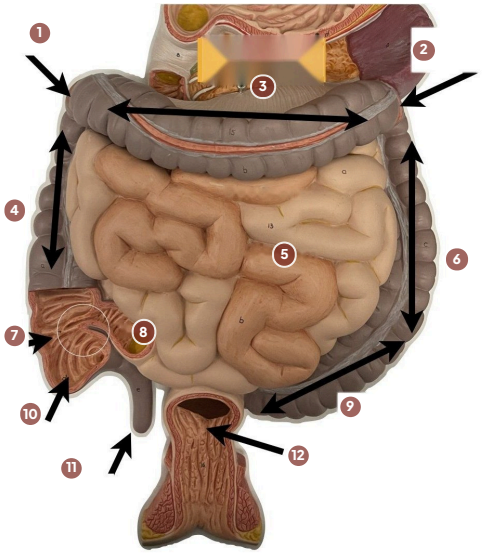
5 _____
Def _____
Location _____
Borders _____
Blood supply (info only) .....

6 _____
Def _____
Function _____

Digestive System, Part III

Competencies 116 to 119: small intestine, large intestine and rectum, the path of food, and the canal disorders. Name _____

A. Small and large intestine in place (model) COMPETENCY 116, 117



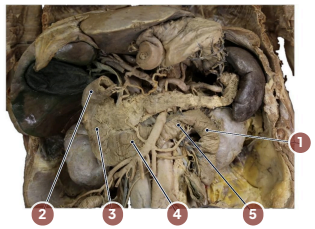
Model, small and large intestine

1 _____ 7 _____
2 _____ 8 _____
3 _____ 9 _____
4 _____ 10 _____
5 _____ 11 _____
6 _____ 12 _____

FILL IN

Duodenum: _____
Jejunum: _____
Ileum, the longest: _____
Food leaves the ileum into the _____
Four parts of the colon: _____

B. Duodenum (cadaver) and the colon surface (model) COMPETENCY 116, 117

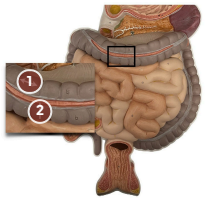


Cadaver, the four parts of the duodenum

FILL IN

Folding, big to small: _____
Lacteal absorbs _____
Large intestine only: _____
Teniae coli are the _____
No haustra on the _____

1 _____
2 _____
3 _____
4 _____
5 _____



1 _____
2 _____

Model, large intestine surface

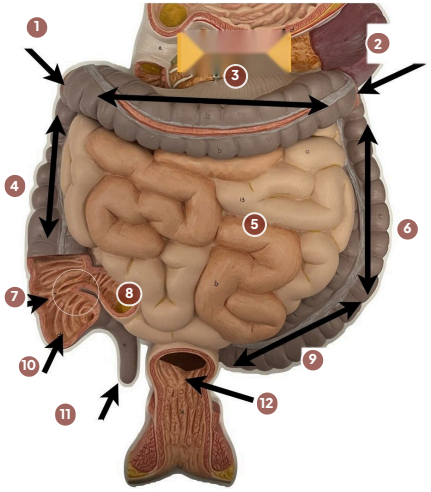
Extra notes and drawings

Extra notes and drawings

Digestive System, Part III STRUCTURE NOTES

A. Small and large intestine in place (model) · Competency 116, 117 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Model, small and large intestine

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

5 _____
Def _____
Function _____

6 _____
Def _____
Function _____

7 _____
Def _____
Function _____

8 _____
Def _____
Function _____

9 _____
Def _____
Function _____

10 _____
Def _____
Function _____

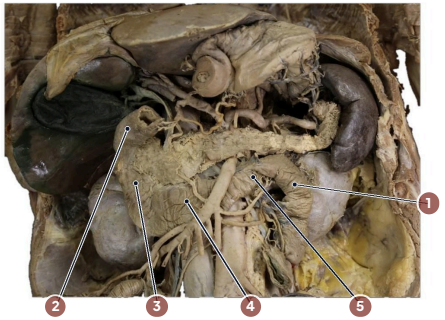
11 _____
Def _____
Function _____

12 _____
Def _____
Function _____

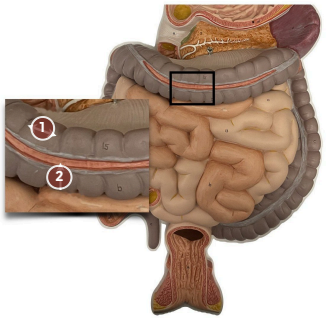
Digestive System, Part III STRUCTURE NOTES

B. Duodenum (cadaver) and the colon surface (model) · Competency 116, 117 · one card per numbered structure, same lines as the slide.
Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Cadaver, the four parts of the duodenum



Model, large intestine surface

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

3 _____
Def _____
Function _____

4 _____
Def _____
Function _____

5 _____
Def _____
Function _____

1 _____
Def _____
Function _____

2 _____
Def _____
Function _____

Digestive System, Part III

Competencies 116 to 119: small intestine, large intestine and rectum, the path of food, and the canal disorders. Name _____

C. Trace a bite of food, mouth to anus COMPETENCY 118

IN ORDER, WITH A RING AT EACH SPHINCTER

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

FILL IN

Not on the path: _____

Most absorption happens in the _____

Water is absorbed and feces form in the _____

Extra notes and drawings

D. Disorders: which structure does each one affect? COMPETENCY 119

DISORDER: STRUCTURE, WHAT CHANGES

GERD: _____

Hiatal hernia: _____

Peptic ulcer: _____

Appendicitis: _____

Diverticulosis: _____

IBD: _____

Hemorrhoids: _____

Colorectal cancer: _____

DRAW: THE FOUR WALL LAYERS, WITH AN ULCER ERODING FROM THE MUCOSA OUTWARD

Extra notes and drawings

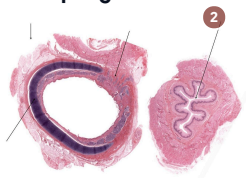
Dr. Sharilyn Rennie · BIO 004 Human Anatomy · Solano Community College · Photos: BIO 004 Digital Atlas, drsrennie-stack.github.io/new-digital-atlas

Digestive System: Histology

The tube and the accessory organs under the microscope. Competencies 109, 113, 115 to 117, 121, 122, 124.

Name _____

A. Esophagus and stomach on the slide COMPETENCY 109, 113, 115



Slide, esophagus (right) beside the trachea

2 _____

Slide photo to add

Slide: stomach wall, gastric pits and glands

FILL IN

Esophagus epithelium: _____

Esophagus outer layer: _____

Stomach surface epithelium: _____

Gastric pits lead to _____

Chief cells make _____

Parietal cells make _____

Stomach muscularis: _____

B. Small intestine and large intestine on the slide COMPETENCY 116, 117

Slide photo to add

Slide: small intestine, villi and crypts

Slide photo to add

Slide: colon, crypts with goblet cells, no villi

FILL IN

Villi are _____

Microvilli form the _____

Crypts (intestinal glands) hold _____

Duodenum only, in the submucosa: _____

Ileum only: _____

Colon mucosa: _____

Colon outer muscle: _____

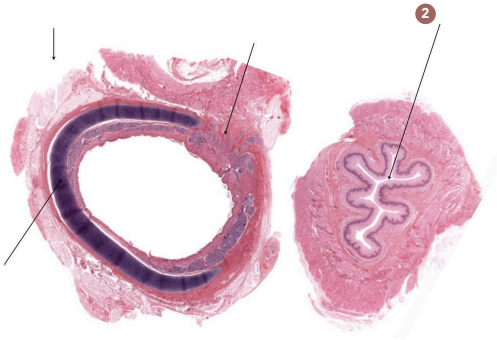
Extra notes and drawings

Extra notes and drawings

Digestive System: Histology STRUCTURE NOTES

A. Esophagus and stomach on the slide · Competency 109, 113, 115 · one card per numbered structure, same lines as the slide. Write the name, fill each line as we go; muscles show the bones with origin (rust) and insertion (blue) shaded.

Name _____



Slide, esophagus (right) beside the trachea

2 _____

Def _____

Location _____

Lining _____

Nerve _____

Blood supply (info only) .....

Function _____

Clinical _____

Digestive System: Histology

The tube and the accessory organs under the microscope. Competencies 109, 113, 115 to 117, 121, 122, 124.

Name _____

C. Liver and pancreas on the slide COMPETENCY 122, 124

Slide photo to add

Slide: liver lobule, central vein and portal triad

Slide photo to add

Slide: pancreas, acini with a pale islet

FILL IN

Six-sided unit: _____

In the center: _____

At the corners: _____

Between the plates: _____

Blood flows _____

Bile flows _____

Pancreas: dark clusters are _____

Pale round patches are _____

Extra notes and drawings

D. Salivary gland on the slide, and the four-layer rule COMPETENCY 109, 121

Slide photo to add

Slide: salivary gland, serous and mucous cells

FILL IN

Serous cells make _____

Mucous cells make _____

Parotid is _____

Sublingual is _____

FOUR LAYERS, ANY SLIDE OF THE TUBE

1. _____
2. _____
3. _____
4. _____

DRAW: A WEDGE OF GUT WALL AS IT LOOKS ON A SLIDE, LUMEN AT THE BOTTOM, THE TWO PLEXUSES DOTTED IN

Extra notes and drawings